

09 What's Broken

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13 MIN READ · INTERACTIVE

Scenes do not fail all at once, and different systems fail under different contracts. A synthesis of what to test, which benchmark claims compose, and which do not.

Every chapter so far has ended with something that does not work yet.

This one collects them, because the gap between what these systems can do and what the announcements imply is where most of the confusion in this subject lives, and because knowing what is broken is the difference between reading a paper and being sold one.

Rollouts do not fail all at once

An average error over a rollout collapses distinct failures into one number. Identity can fail while texture remains sharp; action fidelity can fail while both look fine. The order depends on the system and task, so the useful report is not one horizon but a set of property-specific horizons.

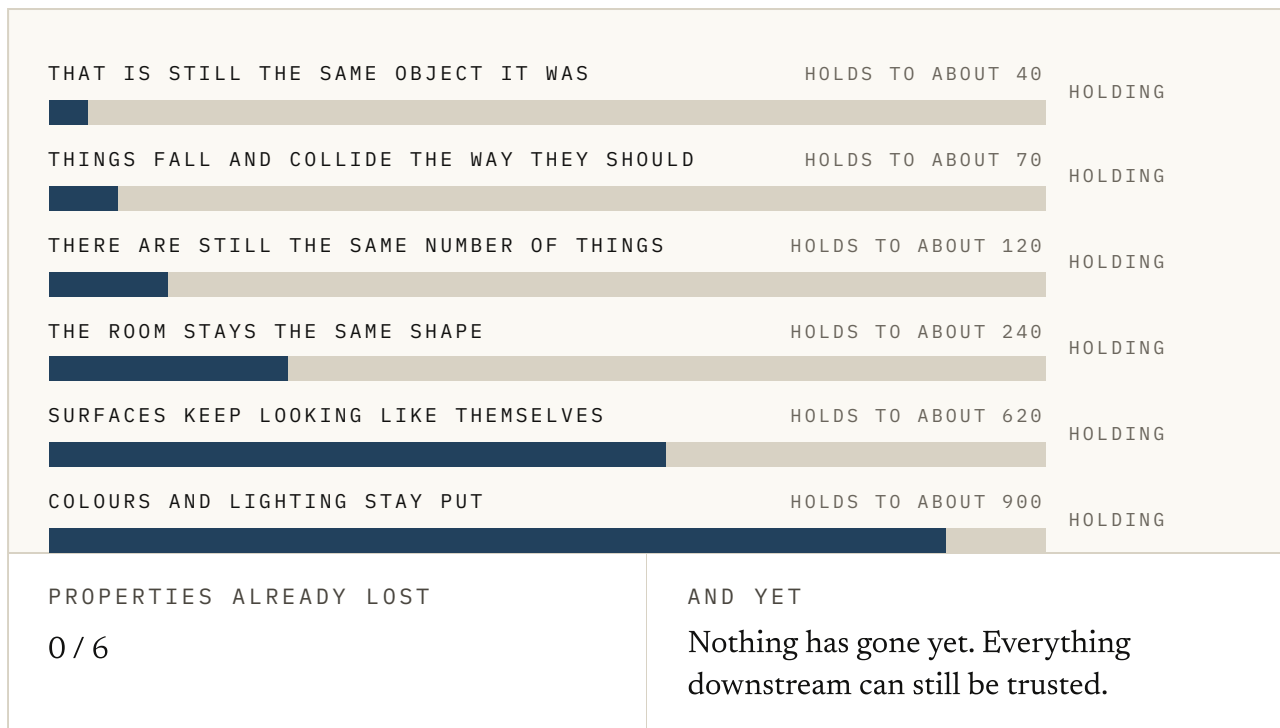


FIG. 9.1 A diagnostic thought experiment, not a benchmark result. Its ordering is one plausible failure profile chosen to expose what an average hides. A different model may lose physics before identity or preserve layout longer. Drag the horizon. "Still looks good" is not the same claim as "still supports the task."

In this constructed profile, colour and texture outlast identity and physical behaviour. That ordering is not universal. Per-frame losses directly reward surfaces and light, while persistence has to be inferred across time.

Which is precisely backwards from what you want, and it is not a coincidence. A loss computed per frame rewards getting most of the pixels approximately right, and most of the pixels are surfaces and light. Nothing in that loss is watching whether the chair you walked past is the same chair.

So a rollout can be a hundred steps past the point of being useful and still look, in any single frame, completely fine. That is the specific way this subject is easy to fool yourself about.

The three that keep coming back

Things stop being themselves. Object permanence is the one everyone notices. In a renderer with no exposed object store, identity is carried by the predictor's hidden state rather than enforced by a persistent record. It can degrade gradually and without an error message when the chair becomes another chair. Structured simulators may store the object explicitly; they pay for that advantage by requiring the structure to be defined or learned correctly.

What if is not really available. A model can tell you what is likely to happen next. Asking what would have happened if you had done something else is a different question, and the answer you get is true inside the model rather than about the world. Where the model has data the two are close. Where it does not, they are not, and nothing in the answer indicates which case you are in.

This one is easy to under-rate because the model always answers. A system that said "I have never seen anything like that" would be more useful and less impressive, and nobody is rewarded for building it.

Long horizons are still the binding constraint. Every trick in this course is, read uncharitably, a way of not needing the model to be right for very long. Short rollouts. Replanning. Starting from real states. They work. They also amount to an admission that the thing does not hold up, and the horizon over which it does is the number that decides what you can build.

There is no agreed overall winner

This is the part that should bother you most, and it gets the least attention.

Benchmarks now test generated worlds directly. PlayWorld, released in August 2026, evaluates 171 scenarios across geometry, interaction fidelity and what happens to objects as they leave and re-enter view. Other suites emphasise

perceptual video quality. What is still missing is an accepted way to turn these different contracts into one overall ordering.

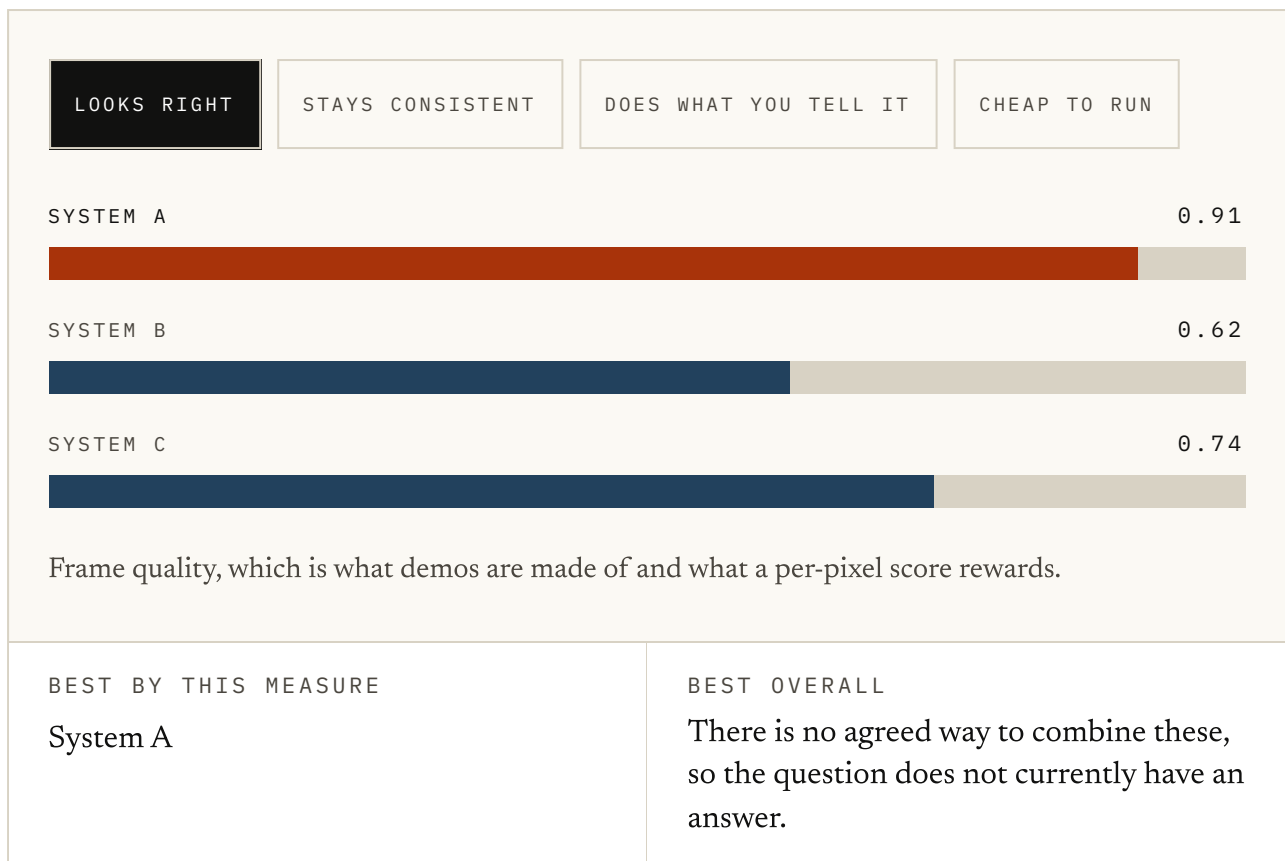


FIG. 9.2 The scores are invented and the systems are not real. The disagreement is real. There is no accepted way to combine these measures into one ordering. Click through the four and watch the winner change.

Frame quality is measurable and is what demos are made of. Whether the world holds together over a thousand steps is what an agent needs, and no demo is long enough to show it. Whether the action you gave it is the action it took is the difference between a world model and a video, and it is almost never reported. Cost decides whether anything can afford to search inside the thing, and it is reported least of all.

Downstream task success looks like the honest answer, and it has its own problem. It scores the model and the policy together, so a good policy hides a bad model and the number tells you nothing about which you have.

Progress is real. "Better" is incomplete until the measure and system contract are named.

Match the failure test to the system

Chapter 1 separated five things that share a name. They should not face the same exam. A renderer lives or dies by persistence and action fidelity. A compact dynamics model lives or dies by rollout horizon and whether planning search can exploit it. A representation can look stable while discarding the variable the task needed. An implicit model needs a probe capable of falsifying the claim that the model is there at all.

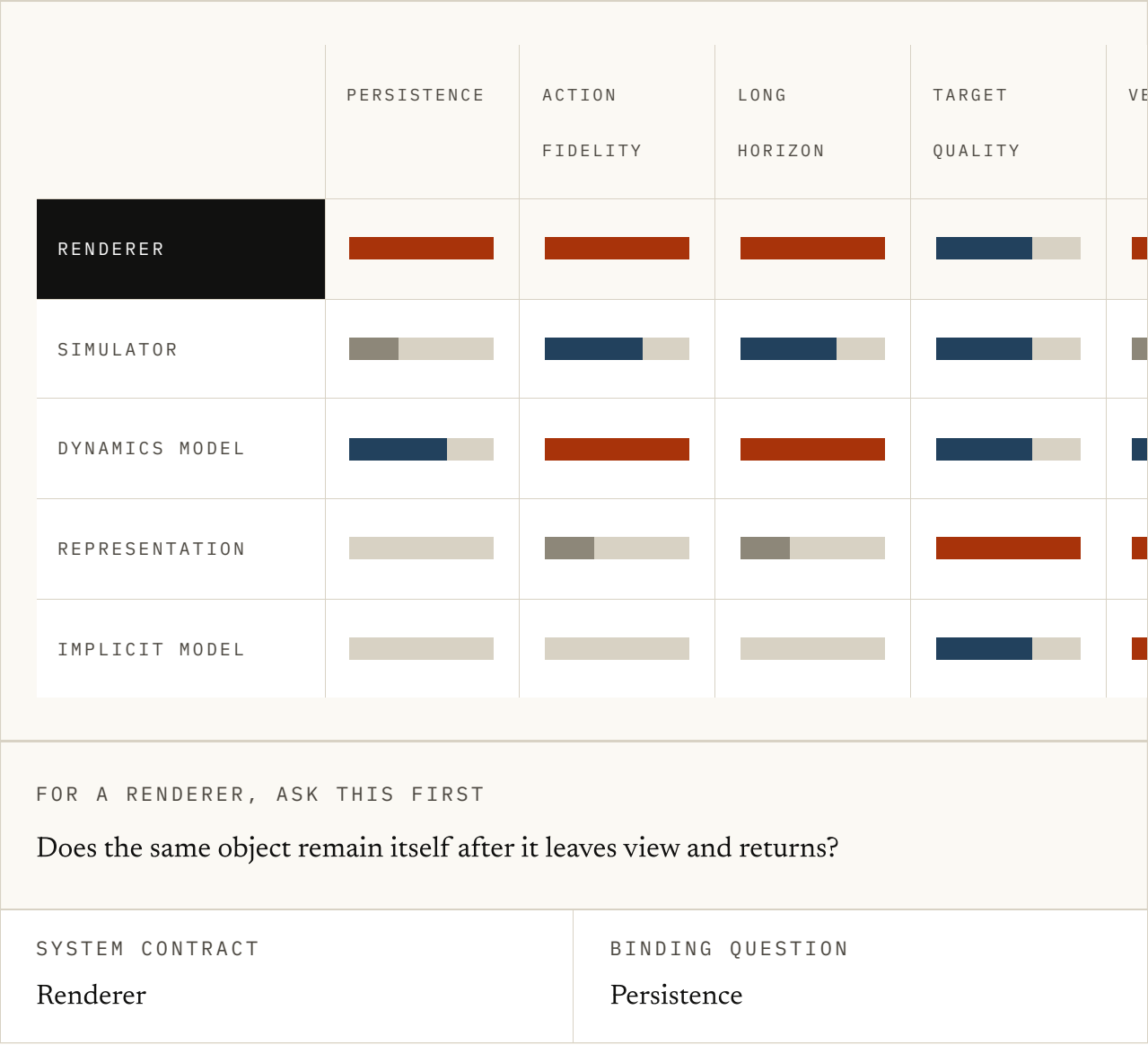


FIG. 9.3 The five Chapter 1 contracts crossed with five failure questions. The levels are editorial judgements, not benchmark scores. Select a row to reveal the question that most often binds that kind of system. A single "world model quality" axis would hide all five.

Four questions still handle most papers:

What does it output? A picture, a structure, a compact state, an embedding. This decides what it can be used for and almost nothing else follows without it.

Can you roll it forward under actions nobody took? If not, it is not going to support a decision, however good it looks.

Over what horizon does it hold up, and who measured that? The number matters more than the architecture, and it is usually missing.

Which measure is the claim using, and what does that measure ignore? There is no overall score. Anyone implying there is has picked one and not told you.

Those questions force a paper to name its output, its evidence and the horizon over which its claim holds. Architecture names alone do none of that work.

Try it

1 Why is one average error insufficient as a long-rollout report?

- a. Different properties can fail at different horizons, and an average hides which contract was lost
- b. The error is measured in the wrong units
- c. Rollouts are too short to average over
- d. Averages are always misleading

ANSWER a. Different properties can fail at different horizons, and an average hides which contract was lost. There is no universal failure order. The point is to report separate horizons for identity, physics, control and appearance instead of allowing one strong dimension to conceal another.

2 A rollout still looks completely fine in any single frame. What does that tell you?

- a. The horizon has not been reached
- b. It is still usable
- c. Very little. Most of the pixels are surfaces and light, and nothing in a per-frame loss is watching whether objects stayed themselves
- d. The model has learned physics

ANSWER c. Very little. Most of the pixels are surfaces and light, and nothing in a per-frame loss is watching whether objects stayed themselves. This is the specific way the subject is easy to fool yourself about, and it is why demos are weaker evidence than they feel.

3 Why is object permanence fragile in a renderer with no exposed object store?

- a. Occlusion is computationally expensive
- b. Training data lacks occluded objects
- c. The models are too small
- d. Identity is carried implicitly by predictor state rather than enforced by a persistent record

ANSWER d. Identity is carried implicitly by predictor state rather than enforced by a persistent record. A structured simulator may store objects explicitly. In the renderer case, identity can drift gradually because persistence is an inferred behaviour rather than an inspectable record.

4 You ask a model what would have happened if you had acted differently. What have you got?

- a. A fact about the world
- b. An answer that is true inside the model, with nothing in it indicating whether the model had data there
- c. A proof
- d. Nothing, models cannot answer that

ANSWER b. An answer that is true inside the model, with nothing in it indicating whether the model had data there. Where the model has data the two are close. Where it does not, they are not, and the model answers with equal confidence either way.

5 Short rollouts, replanning and starting from real states all work. What do they have in common?

- a. They are all ways of not needing the model to be right for very long
- b. They only apply to robotics
- c. They remove the need for a policy
- d. They make the model more accurate

ANSWER a. They are all ways of not needing the model to be right for very long. Read uncharitably, the whole toolkit is an admission. The horizon over which a model holds up is the number that decides what you can build.

6 Why does downstream task success not settle which model is better?

- a. It only works in simulation
- b. Tasks are too easy
- c. It scores the model and the policy together, so a good policy hides a bad model
- d. It cannot be measured reliably

ANSWER c. It scores the model and the policy together, so a good policy hides a bad model. It looks like the honest answer and it conflates the two things you were trying to tell apart.

7 A benchmark winner is best on geometry but mediocre on action fidelity. What is the first thing to ask?

- a. What hardware it runs on
- b. How long it trained for
- c. How large is it
- d. Which measure, and what does that measure ignore

ANSWER d. Which measure, and what does that measure ignore. Benchmarks such as PlayWorld make useful dimensions explicit. They do not remove the need to say which contract matters for the intended use or how competing dimensions were combined.

8 Across the whole course, which single question does the most work on an unfamiliar system?

- a. How many parameters does it have
- b. What does it output, and can you roll it forward under actions nobody took
- c. Is it open source
- d. Does it use a transformer

ANSWER b. What does it output, and can you roll it forward under actions nobody took. The first half decides what it can be used for. The second decides whether it can support a decision. Almost everything else follows, and neither depends on knowing the architecture.

FIG. 9.4 Eight questions on the honest state of it. If you have read the whole course, the last one should be easy and slightly annoying.

That is the course.

Nine chapters ago the phrase meant five different things and nobody said which. It still does. What has changed is that you can now tell which one is on the table, and what it would take for the claim to be true. And what to ask when somebody is not saying.

Thanks for reading. If something here is wrong, and some of it will be, the correction is genuinely welcome and will be credited.

Sources

PlayWorld: A Benchmark for Interactive World Models ZHANG ET AL., 2026 ↗

A current benchmark spanning 171 scenarios and separating geometry, interaction fidelity, and out-of-sight evolution rather than pretending world quality is one number.

<https://arxiv.org/abs/2608.13552>

How Far is Video Generation from World Model: A Physical Law Perspective

KANG ET AL., 2024 ↗

Physical laws matched inside the training distribution and not extrapolated outside it. The clearest measured statement of the gap this chapter is about.

<https://arxiv.org/abs/2411.02385>

VBench: Comprehensive Benchmark Suite for Video Generative Models

HUANG ET AL., 2023 ↗

A serious attempt at the measurement problem, and useful for seeing how many separate dimensions it takes before the ordering stops being obvious.

<https://arxiv.org/abs/2311.17982>

Benchmarking Model-Based Reinforcement Learning WANG ET AL., 2019 ↗

Where model-based methods win and lose, and the discovery that the planning horizon is the number that decides it.

<https://arxiv.org/abs/1907.02057>

When to Trust Your Model: Model-Based Policy Optimisation JANNER ET AL., 2019 ↗

Short rollouts as an answer to a model you cannot trust for long. The title is this chapter's question.

<https://arxiv.org/abs/1906.08253>

Emergent World Representations LI ET AL., 2022 ↗

The strongest available evidence that something structured forms inside a predictor, and a good example of what it takes to show it rather than assert it.

<https://arxiv.org/abs/2210.13382>

Genie 3 GOOGLE DEEPMIND, 2025 ↗

Reported coherence over minutes, from the lab that built it. Included as an example of exactly the kind of claim this chapter is asking you to read carefully.

<https://deepmind.google/blog/genie-3-a-new-frontier-for-world-models/>

A Path Towards Autonomous Machine Intelligence LECUN, 2022 ↗

The most complete statement of what a world model would have to do to be worth the name, which doubles as a list of what is still missing.

<https://openreview.net/forum?id=BZ5a1r-kVsf>

FIG. 9.5 Ordered for someone building on this rather than for historical completeness.